



Sinkhole Susceptibility Mapping in Central Florida

A Machine Learning Approach to Proactive Geohazard Prediction

CAP 5937 — Final Project | Spring 2026

Osama Altarawneh | Bassam Nasir | Muhammed Jaseem P A | Kishan Dhakal

The Problem

- Florida's limestone bedrock dissolves under acidic groundwater, forming underground cavities that collapse without warning
- Central Florida is one of the most sinkhole-prone regions in the United States
- Conventional methods (borehole drilling, ground-penetrating radar) are reactive — triggered only after visible surface damage
- Coverage is spatially limited and expensive at regional scale
- Goal: flip the workflow from reactive response to proactive, region-wide risk prediction

6

Counties
Studied

5,759

Labeled
Samples

13

Conditioning
Factors

Who Uses This and Why

Planning Agencies

Flag high-risk parcels during
permit review before construction

FDOT

Screen roadway corridors before
resurfacing or construction contracts

Insurance Underwriters

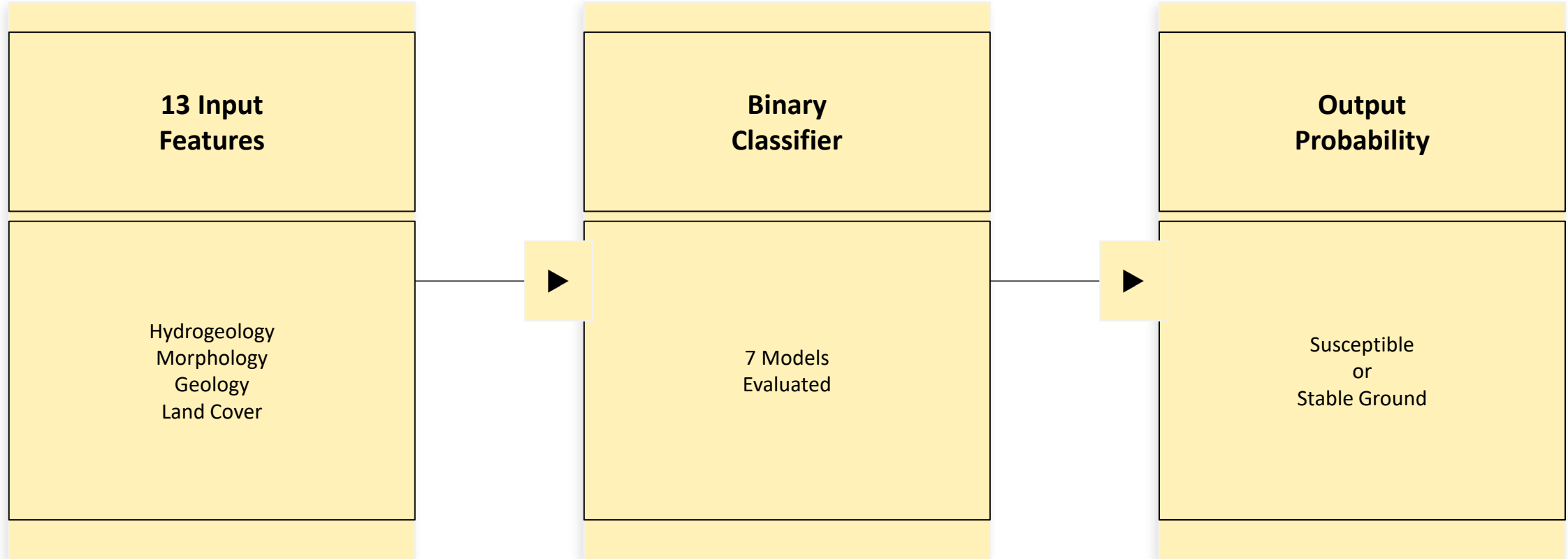
Calibrate sinkhole coverage
pricing using susceptibility scores

Emergency Managers

Prioritize post-storm inspections
during elevated cover-collapse risk

Common benefit: shift from reactive response to spatially targeted, proactive prevention

The ML Problem



Supervised binary classification — each location represented as a 13-dimensional feature vector, output is $P(\text{susceptible})$

Data Collection

- **Positive samples (n = 2,900)**
 - Confirmed sinkhole events from the Florida Geological Survey (FGS) Subsidence Incident Report database to verified cover-collapse and cover-subsidence across 6 counties
- **500 m exclusion buffer**
 - Prevents label noise to a point too close to a known collapse may itself be an unrecorded event mislabeled as stable
- **5,000 m outer boundary**
 - Prevents covariate shift to negative samples share the same hydrogeological context, not geologically dissimilar terrain
- **Near-perfect class balance**
 - 50.4% positive / 49.6% negative across 5,759 samples so no class weighting corrections required

13 Conditioning Features

Hydrogeology

Hydraulic Head
Difference

Intermediate
Aquifer System
(IAS) Thickness

Surficial Aquifer
System (SAS)
Thickness

Surface & Subsurface Morphology

Elevation

Slope

Topographic
Position Index (TPI)

Overburden
Thickness

Shear Wave Velocity

Geological Proximity

Distance to Nearest
Karst Feature

Distance to Nearest
Water Body

Surface Geology
Class

Land Cover

Land Use and Land
Cover (LULC) Class

Mean Annual
Precipitation

Model Results

Random Forest: AUC-ROC 0.891, Accuracy 80.56%, F1 81.14%

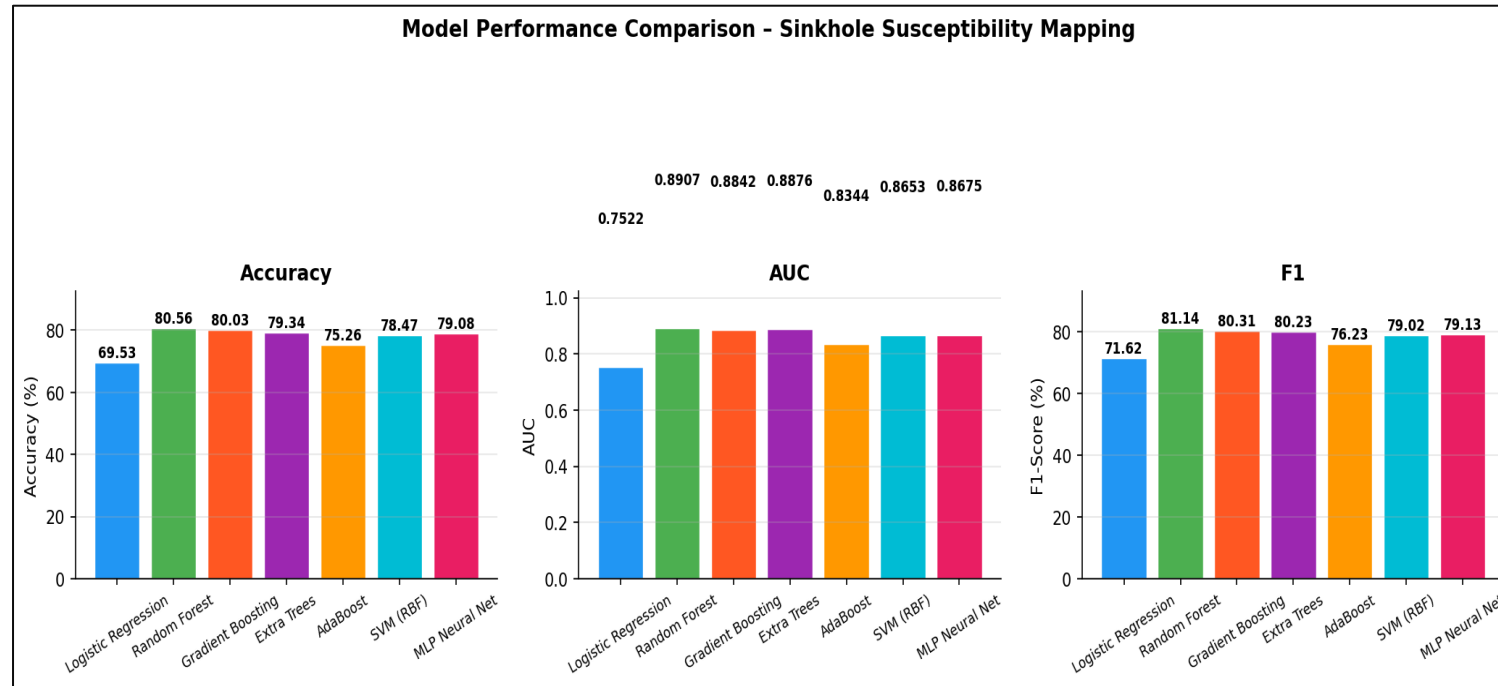


Figure 1: Performance Bar Chart (Accuracy, AUC-ROC, F1 — all 7 models)

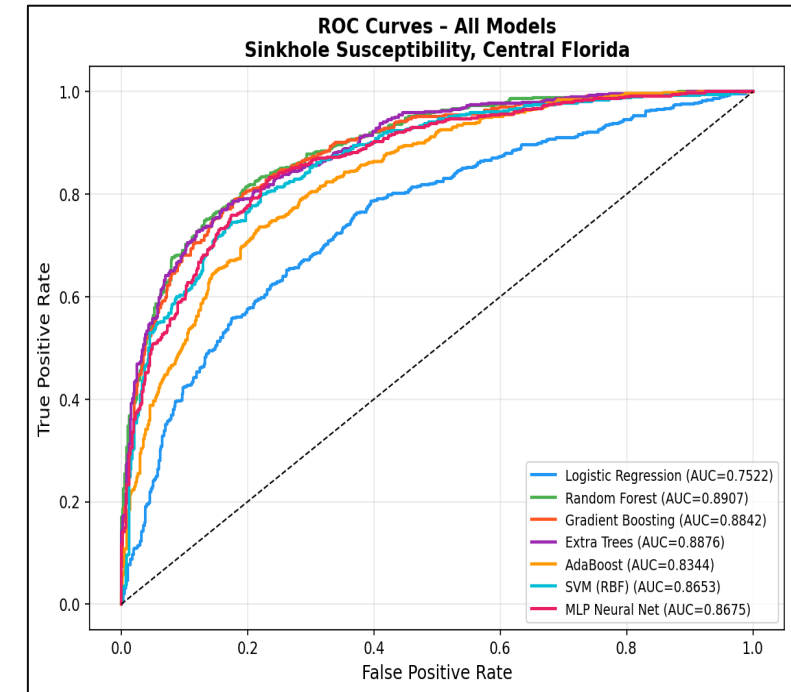


Figure 2: ROC Curves All 7 Models

RF, Extra Trees & Gradient Boosting are statistically indistinguishable (overlapping 95% CIs via Hanley-McNeil). Both are significantly superior to Logistic Regression and AdaBoost.

Logistic Regression linear baseline confirms strong non-linear structure in the problem

Feature Importance

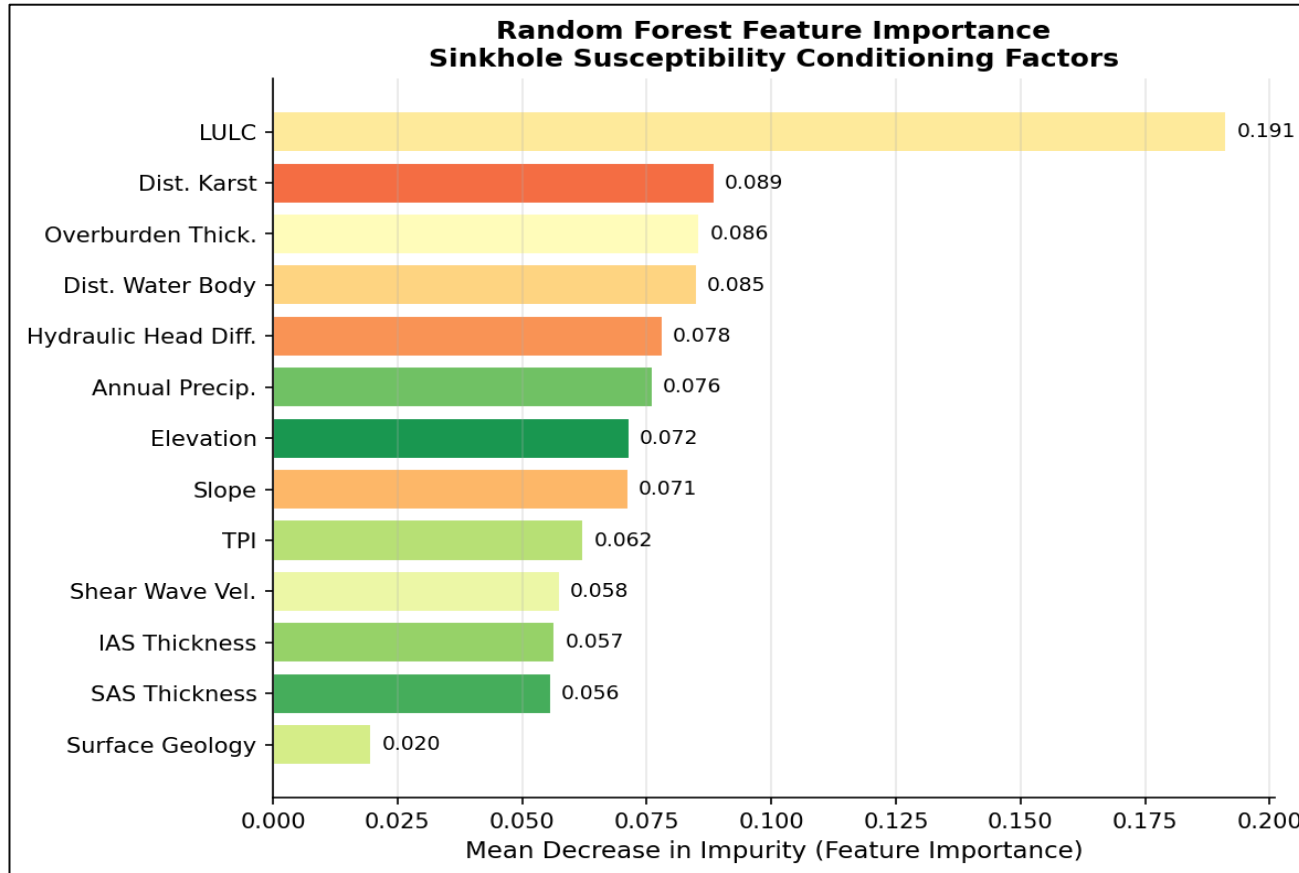


Figure 3 :Random Forest Feature Importance (MDI)

5 Features (RF MDI)

19.1%	LULC Class
8.9%	Distance to Karst Feature
8.6%	Overburden Thickness
8.5%	Distance to Water Body
7.8%	Hydraulic Head Difference

Rankings cross-validated against Gradient Boosting MDI to account for impurity bias toward continuous features

Operational Interpretation

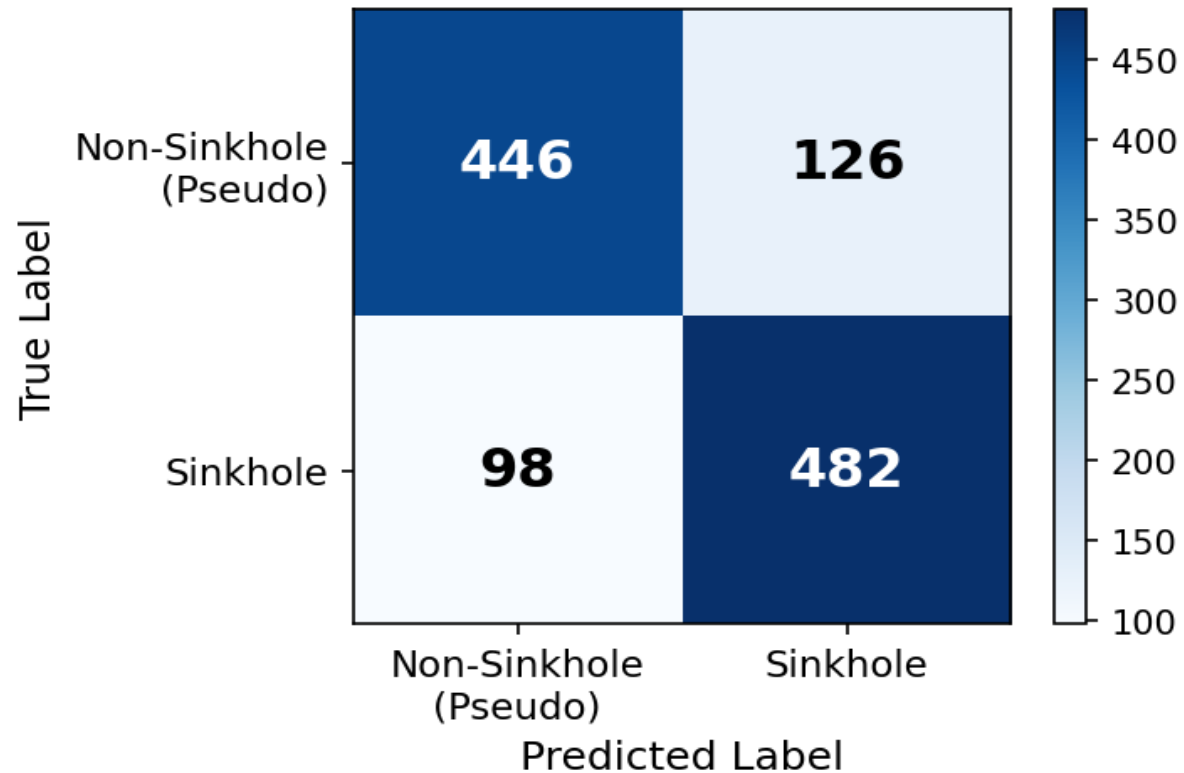


Figure 4 : Confusion Matrix Random Forest (Test set, n = 1,152)

Sensitivity: **83.1%**

831 of every 1,000 true sinkhole locations correctly flagged

Specificity: **78.0%**

220 of every 1,000 stable locations generate an unnecessary alert

Why this trade-off?

Missing a real sinkhole carries far greater cost than an unnecessary inspection
— sensitivity is deliberately prioritized

Conclusions & Limitations

Key Takeaways

- Random Forest (AUC-ROC 0.891) is the best model — ensemble methods consistently outperform linear and shallow boosting approaches
- Spatially controlled pseudo-absence protocol successfully controlled for label noise and covariate shift
- 13 conditioning features carry genuine predictive signal across all seven model families (all AUC > 0.75)
- Statistical significance testing confirms the top 3 ensembles are equivalent; both significantly outperform Logistic Regression and AdaBoost

Limitations & Future Work

- Random train-test split (not spatially stratified) may slightly overestimate generalization to independent regions
- Pseudo-absence labels carry uncertainty — absence of a record is not confirmed ground stability
- Temporal mismatch between historical sinkhole records and current GIS features is unquantified
- Future work: spatial cross-validation, time-matched feature extraction, SHAP-based explainability



Thank you

Osama Altarawneh | Bassam Nasir | Muhammed Jaseem P A | Kishan Dhakal